

An Insight into Advanced Technology in Circulating Fluidised Bed Combustion Steam Generators

The [original paper](#) contains 60 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

Fluidised bed combustion technologies also provide methods for making combustion temperature an independent variable, rather than a consequence of stoichiometry and combustion air temperature. During the combustion process, acid gas control within the furnace by means of direct introduction of sorbent materials into the combustor. These technical advances have caused fluidised bed combustion to become the technology of choice among system designers.

SUMMARY PASSAGE 2

Recycle Ratios

Even though the gas velocity is above the entrainment velocity of most particles in the bed, the entire bed is not entrained out of the combustor. This is because the particles tend to form "clusters" which break-up, move up and down within the combustor. The gas velocity is below the entrainment velocity of these clusters.

SUMMARY PASSAGE 3

Combustor Surface To Volume Ratio

As CFB SGs get larger in size, the combustor surface to volume ratio decreases and it is not possible to perform the required evaporator duty in the furnace water walls enclosure. For example, the CFB SG furnace heights of a 125 MW and 250 MW unit are 33 and 37 metres respectively. The additional duty requirements must be carried out using either in-furnace surfaces or external heat exchanger surfaces.

SUMMARY PASSAGE 4

Turndown

Turndown is accomplished by reducing both fuel and air to the unit. In the process, grate and combustor velocity should be kept above a minimum level in order to produce adequate mixing and solids recirculation for reasonable fuel combustion and to avoid severe temperature maldistribution and back sifting of bed material into the air plenum. This usually implies excess air levels holding constant as load is decreased until the minimum velocity is reached, after which excess air percentage increases to maintain fluidisation velocity.

SUMMARY PASSAGE 5

Fuel Feeding System

The solid fuel feed system usually consists of a pressurised belt feeder followed by a rotary air lock valve and fuel pipe leading to the side of the lower combustor. Fuel from the feeder falls by gravity through the air lock valve into the combustor. The feeder is pressurised.

SUMMARY PASSAGE 6

Combustion Air Supply

On increasing the load, the air rate is the load parameter with coal feed rate being controlled as a function of the air feed rate. On decreasing load, the coal feed rate controller takes the lead and the air feed rate is controlled as a function of the coal feed rate. The delivery head generated by all fans and blowers is sufficient to overcome the pressure losses across the system and the backpressure in the CFB combustor.

SUMMARY PASSAGE 7

Upper Combustor

The upper combustor, section above the refractory lined lower combustor, contains the gas outlets or outlets to cyclones. The density of the bed is relatively low and drops off with increasing combustor height. Because all air has been fed in lower combustor, the upper combustor region operates under excess air (oxidising) conditions.

SUMMARY PASSAGE 8

Fluidised Bed Heat Exchangers

The FBHE is an alternate means used by some of the manufacturers to remove heat from hot solids in the primary solid recirculation loop and reduce combustor size. The FBHE is a bubbling bed heat exchanger consisting of one or more compartments separated by weirs and containing immersed tube bundles. Hot solids from the seal pot enter the FBHE, where they are fluidised and transfer heat to the heating surface within and then flow back to the combustor. The tube bundles immersed in the FBHE compartments can be evaporator, superheater or reheater surface.

SUMMARY PASSAGE 9

Bed Ash System

Bed ash system can also help in controlling accumulation of oversize material. In CFB, such an accumulation can produce an unfavourable pressure profile with most material in the lower combustor and little in the upper combustor resulting in poor performance. However the best and most direct way to control oversize accumulation is with proper design of fuel sizing equipment to avoid oversize.

SUMMARY PASSAGE 10

Ash Classifiers

Such classifiers can operate continuously or in batch mode and can also cool the ash. Bottom ash must be cooled from combustor temperature to around 120 -150 deg C, before entering bottom ash conveying system. On high-ash fuels, the heat from the bottom ash stream may represent a significant percentage of heat.